

COMPUTER FUNDAMENTALS

Unit-III: Input and Output Devices

Comprehensive Study Notes

1. Input Devices

1.1 What are Input Devices?

Input devices are hardware components used to **enter data and instructions** into a computer system. They convert user actions or external data into a format that the computer can understand and process. Without input devices, users cannot interact with the computer or provide it with the information needed to perform tasks. Input devices are the primary means of communication between the user and the computer system.

Key Functions of Input Devices

- Accept data from the user or environment
- Convert data into electrical signals
- Send converted data to the computer for processing

1.2 Classification of Input Devices

Input devices can be classified into several categories based on their functionality and the type of data they handle:

Category	Devices
Keyboard	Standard, Wireless, Gaming, Ergonomic
Pointing	Mouse, Touchscreen, Joystick, Trackball, Stylus
Scanning	Scanner, OCR, OMR, Bar Code Reader, MICR, Digitizer
Image Capturing	Digital Camera, Webcam

2. Keyboard

2.1 What is a Keyboard?

A keyboard is the **most common input device** used to enter text, numbers, and commands into a computer. It consists of a set of keys arranged in a specific layout, typically following the QWERTY pattern.

2.2 Keyboard Layout

QWERTY Layout: The standard keyboard layout named after the first six letters in the top row: Q, W, E, R, T, Y.

Parts of a Keyboard

Part	Description
Alphanumeric Keys	Letters (A-Z) and numbers (0-9)
Function Keys	F1 to F12 — perform specific functions
Navigation Keys	Arrow keys, Home, End, Page Up, Page Down
Numeric Keypad	Calculator-like keys for numbers
Modifier Keys	Shift, Ctrl, Alt, Caps Lock
Special Keys	Enter, Backspace, Delete, Tab, Esc

2.3 Types of Keyboards

Type	Description	Usage
Mechanical Keyboard	Uses mechanical switches, tactile feedback	Gaming, heavy typing

Membrane Keyboard	Uses pressure pads, quiet, cheap	General use, offices
Wireless Keyboard	Connects via Bluetooth or USB receiver	Laptops, portable devices
Virtual Keyboard	On-screen touch keyboard	Tablets, smartphones
Gaming Keyboard	RGB lights, macro keys, fast response	Gaming
Ergonomic Keyboard	Split design, reduces wrist strain	Long typing sessions

2.4 Advantages of Keyboard

- Fastest text input device
- Easy to learn and use
- Supports multiple languages
- Provides tactile feedback
- Can perform shortcuts (Ctrl+C, Ctrl+V)

2.5 Disadvantages of Keyboard

- Requires training for fast typing
- Can cause repetitive strain injuries
- Not suitable for graphical input
- Physical space required

3. Pointing Devices

3.1 What are Pointing Devices?

Pointing devices are input devices that allow users to control the **movement of a cursor** or pointer on the screen. They are essential for graphical user interfaces (GUI) and navigation.

3.2 Mouse

What is a Mouse?

A mouse is the most common pointing device. It detects two-dimensional motion relative to a surface and translates it into cursor movement on the screen.

Types of Mouse

Type	Description	Technology
Mechanical Mouse	Uses a rubber ball to track movement	Ball movement
Optical Mouse	Uses LED light to detect surface	Optical sensor
Laser Mouse	Uses laser for precise tracking	Laser sensor
Wireless Mouse	Connects without wires	Bluetooth/RF
Gaming Mouse	High DPI, extra buttons, RGB	Advanced sensor

How a Mouse Works

1. User moves the mouse on a surface
2. The tracking mechanism detects movement
3. Movement is converted into electrical signals
4. Signals are sent to the computer
5. Cursor moves accordingly on the screen

Parts of a Mouse

- **Left Button:** Select, click, drag
- **Right Button:** Context menu
- **Scroll Wheel:** Scroll up/down
- **Middle Button:** Additional functions
- **Optical/Laser Sensor:** Tracks movement

Advantages of Mouse

- Easy to use
- Fast navigation
- Precise control
- Ideal for GUIs

Disadvantages of Mouse

- Requires flat surface
- Can cause wrist strain
- Not suitable for all applications

3.3 Touch Screen

What is a Touch Screen?

A touch screen is a **display screen** that can detect the touch of a finger or stylus. It combines the functions of a display and a pointing device into one.

Types of Touch Screen Technologies

Type	Description	Advantages	Disadvantages
Resistive	Two layers separated by air gap	Cheap, works with any object	Less accurate
Capacitive	Uses electrical charge	Multi-touch, clear display	Expensive, needs finger

Infrared	Uses IR beams	Durable, clear	Expensive
Surface Acoustic Wave	Uses sound waves	Very clear, durable	Expensive

Uses of Touch Screens

- Smartphones and tablets
- ATMs and kiosks
- Laptops with touch display
- Industrial control panels
- Point of Sale (POS) systems

3.4 Joystick

What is a Joystick?

A joystick is a pointing device consisting of a **stick that pivots on a base** and reports its angle or direction to the computer.

How a Joystick Works

1. User moves the stick in any direction
2. Sensors detect movement and angle
3. Signals are sent to the computer
4. Cursor or object moves accordingly

Types of Joysticks

Type	Description	Usage
Analog Joystick	Continuous movement detection	Gaming, flight simulators
Digital Joystick	On/Off directional control	Arcade games
Force Feedback Joystick	Provides tactile feedback	Advanced gaming

Advantages of Joystick

- Intuitive for gaming
- Good for flight simulators
- Easy to control direction
- Provides feedback

Disadvantages of Joystick

- Not suitable for typing
- Requires practice
- Limited to specific applications

3.5 Electronic Pen / Stylus

What is an Electronic Pen?

An electronic pen (or stylus) is an input device that **looks like a pen** but is used to interact with touchscreens or tablets.

Types of Electronic Pens

Type	Description
Capacitive Stylus	Works with capacitive touchscreens
Active Stylus	Uses Bluetooth for precise input
Digital Pen	Converts handwriting to digital text

Uses of Electronic Pen

- Drawing and graphic design
- Taking notes on tablets
- Signing documents
- Precise selections on touchscreens

3.6 Trackball

What is a Trackball?

A trackball is a pointing device that consists of a **ball** held in a socket with sensors to detect rotation. The user rotates the ball to move the cursor.

Advantages of Trackball

- Stationary — no need to move the device
- Saves desk space
- Good for users with limited space

Disadvantages of Trackball

- Requires practice
- Not as intuitive as a mouse
- Less common

4. Scanning Devices

4.1 What are Scanning Devices?

Scanning devices are input devices that **convert physical documents, images, or objects** into digital formats that can be stored, processed, or displayed on a computer.

4.2 Optical Scanners

What is an Optical Scanner?

An optical scanner is a device that **captures images** from physical documents and converts them into digital format using light sensors.

How a Scanner Works

1. Document is placed on the scanner bed
2. Light moves across the document
3. Light is reflected from the document
4. CCD sensors capture the reflected light
5. Data is converted to digital image

Types of Scanners

Type	Description	Usage
Flatbed Scanner	Document placed on glass	General scanning
Sheet-fed Scanner	Document pulled through	Multi-page scanning
Drum Scanner	High quality scanning	Professional photography
Handheld Scanner	Manual movement	Quick scanning

4.3 Optical Character Recognition (OCR)

What is OCR?

OCR is a technology that **converts printed or handwritten text** into machine-readable digital text. It allows computers to "read" text from scanned documents.

How OCR Works

1. Image of text is captured (scanned)
2. Software analyzes the image
3. Characters are recognized and isolated
4. Recognized characters are converted to ASCII text
5. Text can be edited, searched, and stored

Applications of OCR

- Document digitization
- Data entry automation
- Book scanning and conversion
- Form processing
- Number plate recognition

4.4 Optical Mark Recognition (OMR)

What is OMR?

OMR is a technology that **detects marks** made on a specific form, such as filled circles or shaded areas. It is commonly used for test scoring and surveys.

How OMR Works

1. Special form with bubbles or marks is created
2. User fills the bubbles with a pencil or pen
3. OMR reader scans the form
4. Light sensors detect filled areas
5. Responses are recorded

Applications of OMR

- Multiple-choice test scoring

- Surveys and questionnaires
- Attendance tracking
- Voting systems
- Feedback forms

4.5 Bar Code Reader

What is a Bar Code Reader?

A bar code reader is an optical device that **reads bar codes** — patterns of parallel lines that represent data.

How a Bar Code Reader Works

1. Laser or LED light is directed at the bar code
2. Light is reflected from the bar code
3. Dark bars absorb light, white bars reflect it
4. Sensor detects the reflected pattern
5. Pattern is decoded into data

Types of Bar Code Readers

Type	Description	Usage
Handheld Reader	Manually aimed at bar code	Retail, warehouses
Stationary Reader	Fixed position reader	Conveyor belts
Smartphone Reader	Uses phone camera	Consumer apps
Pen Reader	Contact scanning	Small scale

Types of Bar Codes

Type	Description	Example
UPC	Universal Product Code	Retail products
Code 39	Alphanumeric bar code	Industrial use

Code 128	High density alphanumeric	Logistics
QR Code	Two-dimensional barcode	Mobile scanning

Applications of Bar Code Readers

- Retail product scanning
- Inventory management
- Library book tracking
- Patient identification in hospitals
- Ticketing systems

4.6 Magnetic Ink Character Recognition (MICR)

What is MICR?

MICR is a technology used in **banking** to read characters printed with special magnetic ink. It is primarily used for processing cheques.

How MICR Works

1. Characters are printed with magnetic ink
2. Document passes through MICR reader
3. Magnetic sensor reads the characters
4. Data is decoded and processed
5. Information is sent to the computer

MICR Characters

- Special font: E-13B or CMC-7
- Characters: 0-9, and special symbols
- Magnetic ink contains iron oxide

Advantages of MICR

- High accuracy
- Fast processing
- Tamper-resistant

- Reliable even if characters are damaged

Applications of MICR

- Cheque processing in banks
- Sorting of bank documents
- Financial transactions

4.7 Digitizer

What is a Digitizer?

A digitizer (or graphics tablet) is a device that **converts analog data (hand-drawn images) into digital format** for use in computers.

How a Digitizer Works

1. User draws on the tablet surface
2. Tablet detects the position of the stylus
3. Position data is sent to the computer
4. Image is created on the screen

Types of Digitizers

Type	Description	Usage
Graphics Tablet	Uses stylus and surface	Graphic design
Digitizing Tablet	Converts drawings to digital	CAD, engineering
Pen Display	Draws directly on screen	Professional design

Applications of Digitizers

- Graphic design and animation
- Photo editing
- Computer-Aided Design (CAD)
- Architectural drawings
- Signature capture

4.8 Electronic Card Reader

What is an Electronic Card Reader?

An electronic card reader is a device that **reads data stored on cards** such as credit cards, ID cards, or access cards.

Types of Card Readers

Type	Description	Usage
Magnetic Stripe Reader	Reads magnetic stripe data	Credit/debit cards
Smart Card Reader	Reads embedded chip	Access control
RFID Reader	Radio frequency identification	Contactless cards
Biometric Reader	Uses fingerprints or retina	High security

Applications of Card Readers

- Payment processing
- Access control
- Identity verification
- Loyalty programs
- E-passport reading

5. Image Capturing Devices

5.1 What are Image Capturing Devices?

Image capturing devices are input devices that **capture visual information** and convert it into digital images or video.

5.2 Digital Cameras

What is a Digital Camera?

A digital camera is a device that **captures photographs** digitally, storing them in a format that can be transferred to a computer.

How a Digital Camera Works

1. Light enters through the lens
2. Light falls on an image sensor (CCD/CMOS)
3. Sensor converts light to electrical signals
4. Analog signals are converted to digital
5. Digital data is stored on memory card

Types of Digital Cameras

Type	Description	Usage
Compact Camera	Small, simple, automatic	Everyday use
Bridge Camera	Between compact and DSLR	Enthusiasts
DSLR	Digital Single Lens Reflex	Professional
Mirrorless Camera	No mirror, compact	Professional
Action Camera	Rugged, wide-angle	Sports, adventure

Key Components of Digital Camera

Component	Function
Lens	Focuses light on sensor
Image Sensor (CCD/CMOS)	Converts light to electrical signal
Processor	Processes image data
Memory Card	Stores photos and videos
LCD Screen	Displays photos and previews

Advantages of Digital Cameras

- Instant preview
- No film processing
- High storage capacity
- Easy to share photos
- Video recording capability

Applications of Digital Cameras

- Professional photography
- Family and personal use
- Security and surveillance
- Scientific documentation
- Medical imaging

5.3 Webcam

What is a Webcam?

A webcam is a video camera that **streams video or captures images** in real time to a computer or network.

Characteristics

- Small and lightweight
- Typically USB connection
- Built-in microphone
- Fixed or adjustable lens

Uses of Webcam

- Video conferencing
- Live streaming
- Security monitoring
- Online teaching
- Social media video

6. Output Devices

6.1 What are Output Devices?

Output devices are hardware components that **display, present, or print** the results of computer processing in a human-readable form.

6.2 Classification of Output Devices

Output devices are classified based on the type of output they produce: visual output (monitors, projectors), hard copy output (printers, plotters), and audio output (speakers).

7. Monitors

7.1 What is a Monitor?

A monitor (or Visual Display Unit — VDU) is an **output device** that displays visual information generated by the computer. It is the most common output device for desktop computers.

7.2 Types of Monitors

CRT (Cathode Ray Tube) Monitor

What It Is: CRT was the traditional monitor technology used in older computers. It uses a vacuum tube with an electron gun to display images.

How It Works:

1. Electron gun fires electrons
2. Electrons strike phosphor coating on screen
3. Phosphor emits light to create image
4. Electron beam scans line by line

Characteristics: Bulky and heavy, high power consumption, good contrast and color, slow refresh rate, **obsolete technology**.

Advantages

Disadvantages

Good color reproduction	Heavy and large
Wide viewing angle	High power consumption
Durable	Flickering can cause eye strain
—	Emits radiation

LCD (Liquid Crystal Display) Monitor

What It Is: LCD monitors use liquid crystals to display images. They are thinner, lighter, and consume less power than CRT monitors.

How It Works:

1. Backlight produces white light
2. Light passes through liquid crystals
3. Liquid crystals twist to control light
4. Color filters create colored image

Characteristics: Thin and lightweight, low power consumption, no flickering, good resolution.

Advantages

Disadvantages

Thin and light	Limited viewing angle
Low power consumption	Can have dead pixels
No flickering	Slower response time
Less eye strain	Backlight bleeding possible
No radiation	—

LED (Light Emitting Diode) Monitor

What It Is: LED monitors are actually LCD monitors that use LED backlighting instead of traditional cold cathode fluorescent lamps (CCFL).

Types of LED Backlighting:

- **Edge-lit:** LEDs along the edges
- **Direct-lit:** LEDs behind the screen
- **Full-array:** LEDs distributed evenly

Advantages over LCD: Better contrast, higher brightness, more energy efficient, thinner design.

OLED (Organic LED) Monitor

What It Is: OLED monitors use organic compounds that emit light when an electric current is applied. Each pixel produces its own light, eliminating the need for backlighting.

Characteristics: Self-emitting pixels, true blacks, excellent contrast, very thin and flexible.

Advantages

Disadvantages

Perfect black levels	Expensive
Infinite contrast ratio	Burn-in risk
Faster response time	Shorter lifespan
Wide viewing angle	—

7.3 Monitor Specifications

Specification

Description

Resolution	Number of pixels (e.g., 1920x1080)
Size	Diagonal measurement in inches
Refresh Rate	Hz — Frames per second
Response Time	ms — Speed of pixel change
Contrast Ratio	Difference between black and white
Brightness	cd/m ² — Light output
Color Depth	Number of colors supported

7.4 Resolution Comparison

Resolution

Pixels

Common Name

640x480	307,200	VGA
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1280x720	921,600	HD Ready
1920x1080	2,073,600	Full HD
2560x1440	3,686,400	QHD
3840x2160	8,294,400	4K UHD

8. Printers

8.1 What is a Printer?

A printer is an output device that **produces a hard copy** (physical copy) of documents, images, or other data on paper or other media.

8.2 Types of Printers

Dot Matrix Printer

What It Is: An impact printer that uses a print head with pins to hit a ribbon against paper, forming characters by dots.

How It Works:

1. Print head moves across the paper
2. Pins strike the ribbon
3. Ribbon creates dots on paper
4. Multiple pins form characters

Characteristics: Impact technology, uses continuous paper, noisy operation, low resolution (72-144 dpi).

Advantages

Disadvantages

Can print on multi-part forms	Noisy
Durable	Low quality output
Low running cost	Slow speed
Works in harsh environments	Limited colors (usually monochrome)

Applications: Receipt printing, invoices, multi-part forms, industrial use.

Inkjet Printer

What It Is: A non-impact printer that uses tiny droplets of ink to create images on paper.

How It Works:

1. Print head moves across the paper
2. Ink is heated or forced through nozzles
3. Tiny droplets are sprayed on paper
4. Droplets form text and images

Types of Inkjet Technology:

- **Thermal Inkjet:** Heat creates bubbles
- **Piezoelectric Inkjet:** Electric charge forces ink

Characteristics: High quality output, affordable, good for color printing, relatively slow.

Advantages	Disadvantages
Good quality	Ink can be expensive
Color printing	Slow for large volumes
Affordable price	Head can clog if not used
Quiet operation	Limited to specific paper types

Applications: Home and small office, photo printing, document printing, graphic design.

Laser Printer

What It Is: A high-speed, high-quality printer that uses laser technology to create images on paper using toner (powdered ink).

How It Works:

1. Laser beam charges the drum
2. Toner sticks to charged areas
3. Drum transfers toner to paper
4. Heat fuses toner to paper

Steps:

1. **Charging:** Drum receives electrical charge
2. **Exposing:** Laser writes image on drum
3. **Developing:** Toner sticks to charged areas
4. **Transferring:** Toner moves to paper
5. **Fusing:** Heat bonds toner to paper

Characteristics: Fast printing speed, high quality output, cost-effective for large volumes, sharp text.

Advantages

Disadvantages

Very fast	Expensive initial cost
High quality	Color lasers are expensive
Low cost per page	Toner cartridges cost more
Sharp text	Larger size
Quiet operation	—

Applications: Office printing, high-volume printing, professional documents, network printing.

8.3 Printer Comparison

Feature	Dot Matrix	Inkjet	Laser
Speed	Slow	Moderate	Fast
Quality	Low	High	Very High
Cost	Low	Low	High
Running Cost	Low	High	Low
Color	Limited	Excellent	Good
Noise	High	Low	Low
Uses	Multi-part	Photos	Office

9. Plotters

9.1 What is a Plotter?

A plotter is a **large-format output device** that produces high-quality, precise drawings and designs. It is used for printing large graphics, maps, blueprints, and engineering drawings.

9.2 Types of Plotters

Drum Plotter

What It Is: A plotter where the paper is placed on a rotating drum while the writing mechanism moves back and forth.

How It Works:

1. Paper is placed on a cylindrical drum
2. Drum rotates to provide Y-axis movement
3. Writing head moves along X-axis
4. Pens create drawings

Advantages: High precision, good for long drawings, continuous paper feed.

Disadvantages: Expensive, large size, limited to roll paper.

Flatbed Plotter

What It Is: A plotter where the paper is placed on a flat surface and the writing mechanism moves over it in both X and Y directions.

How It Works:

1. Paper is placed on a flat table
2. Writing head moves in both directions
3. Pens create drawings

Advantages: Can handle different paper sizes, easy to change paper, good for graphics.

Disadvantages: Slow, expensive, takes more space.

9.3 Plotter vs Printer

Feature	Plotter	Printer
Output	Graphics, drawings	Text, images
Size	Large format	Small/medium
Quality	Very high	Varies
Cost	Expensive	Affordable
Use	CAD, engineering	General purpose

10. Screen Image Projector

10.1 What is a Screen Image Projector?

A screen image projector is an output device that **displays computer images** on a large screen for viewing by multiple people.

10.2 Types of Projectors

Type	Description	Usage
LCD Projector	Uses liquid crystals	Education, business
DLP Projector	Uses digital micro-mirrors	Cinema, home theater
LED Projector	Uses LED light source	Portable, long life
Laser Projector	Uses laser light	High-end, cinema

10.3 How a Projector Works

1. Computer sends video signal
2. Projector processes the signal
3. Light source creates light beam
4. Image is formed using LCD/DLP technology
5. Light projects image onto screen

10.4 Applications of Projectors

- Education and training
- Business presentations
- Cinema and home theater
- Gaming
- Events and conferences

11. Output Devices Comparison

11.1 Output Devices Summary

Device	Output Type	Speed	Quality	Cost
CRT Monitor	Visual	Fast	Good	Low
LCD Monitor	Visual	Fast	High	Medium
Dot Matrix Printer	Hard Copy	Slow	Low	Low
Inkjet Printer	Hard Copy	Medium	High	Medium
Laser Printer	Hard Copy	Fast	Very High	High
Plotter	Hard Copy	Slow	Very High	Very High
Projector	Visual	Fast	Good	High

12. Important Full Forms

Abbreviation	Full Form
CRT	Cathode Ray Tube
LCD	Liquid Crystal Display
LED	Light Emitting Diode
OLED	Organic Light Emitting Diode
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
VDU	Visual Display Unit

Questions

Short Answer Questions (2-5 Marks)

1. What are input devices? Give examples.
2. What is the function of a keyboard?
3. Explain the working of a mouse.
4. What is a touch screen? Name its types.
5. What is a scanner? Name different types.
6. What is OCR? Where is it used?
7. What is OMR? Give its applications.
8. What is a bar code reader?
9. What is MICR? Where is it used?
10. What is the difference between CRT and LCD monitors?
11. Explain the working of a laser printer.
12. What is the difference between inkjet and laser printers?
13. What is a plotter? Name its types.
14. What is a webcam? What are its uses?
15. What is the difference between input and output devices?

Long Answer Questions (10 Marks)

1. What are input devices? Explain different types of input devices in detail.
2. Explain the working and types of keyboards and pointing devices.
3. What are scanning devices? Explain OCR, OMR, MICR, and Bar Code Reader.
4. What are output devices? Explain different types of monitors.
5. Explain the working of different types of printers — Dot Matrix, Inkjet, and Laser.
6. What are plotters? Explain drum and flatbed plotters.
7. Explain the difference between CRT, LCD, LED, and OLED monitors.
8. What is a projector? Explain its types and uses.

Objective Questions (1 Mark)

1. The most common input device is: (a) Monitor (b) Keyboard (c) Printer (d) Speaker

Answer: (b)

2. Which device is used to read bar codes? (a) Scanner (b) Bar Code Reader (c) Printer (d) Plotter

Answer: (b)

3. MICR is used in: (a) Banking (b) Education (c) Retail (d) Hospitals

Answer: (a)

4. Which printer uses toner? (a) Inkjet (b) Dot Matrix (c) Laser (d) None

Answer: (c)

5. CRT stands for: (a) Cathode Ray Tube (b) Computer Ray Tube (c) Cathode Radio Tube (d) None

Answer: (a)

6. Which monitor is most energy efficient? (a) CRT (b) LCD (c) OLED (d) Both b and c

Answer: (d)

7. Which device is used for large format drawings? (a) Printer (b) Plotter (c) Monitor (d) Scanner

Answer: (b)

End of Unit-III Notes